

GATE DA (Data Science & AI) Practice Question Bank

5 Topics · Previous-Year-Style + New Practice Questions · Full Solutions

This document covers five topics: Simple Linear Regression (Cost Function), Ratio & Proportion, Functions/Recursion/Scope Rules, String Manipulation with Two-Pointer technique, and Problem Formulation & State-Space Representation in AI.

IMPORTANT NOTE ON AUTHENTICITY: GATE DA (Data Science & AI) has been conducted only three times so far (2024, 2025, 2026), and it is a non-coding, MCQ/MSQ/NAT written exam -- it never asks candidates to write or submit running code. Because of this, a genuine set of 10 verbatim previous-year questions purely on a narrow sub-topic (e.g. "two-pointer string problems") simply does not exist in the GATE DA archive. To keep this resource honest and useful, every "old" question below is tagged with its real provenance:

[VERIFIED PYQ] -- exact question (or a faithful, fully-worked reconstruction of a well-documented one) from an actual GATE exam, with year and paper noted. Where the source paper is GATE CSE or the shared General Aptitude section rather than GATE DA specifically, this is stated -- these test the identical concept/format used in GATE DA.

[GATE-style] -- not a verbatim previous-year question (either because no verified GATE DA PYQ exists on that exact sub-point, or because reproducing the original figure/data table reliably was not possible), but written to match real GATE DA difficulty, phrasing, and mark-weightage.

Section B in every topic ("New Practice Questions") is, as requested, freshly written GATE-level practice -- none of these are claimed to be past questions.

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Topic 1: Simple Linear Regression - Cost Function

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style -- concept confirmed tested in GATE DA 2025]

Q1. [MCQ · 1 mark]

In simple linear regression **without an intercept**, i.e. $\hat{y} = w \cdot x$, fitted to m data points (x_i, y_i) by minimizing the sum of squared errors, the optimal value of w is:

- (A) $w = \sum x_i y_i / \sum x_i^2$
- (B) $w = \sum x_i y_i / \sum y_i^2$
- (C) $w = (\sum x_i)(\sum y_i) / m$
- (D) $w = \sum x_i^2 / \sum x_i y_i$

Answer: (A)

Solution: $J(w) = \sum (y_i - wx_i)^2$. Setting $dJ/dw = -2\sum x_i(y_i - wx_i) = 0$ gives $\sum x_i y_i = w \sum x_i^2$, so $w = \sum x_i y_i / \sum x_i^2$. (This exact formula-recall question format matches the concept tested in GATE DA 2025.)

[GATE-style]

Q2. [MCQ · 1 mark]

For simple linear regression, the cost function $J(w_0, w_1) = (1/2m) \sum (w_0 + w_1 x_i - y_i)^2$ viewed as a function of the parameters (w_0, w_1) is:

- (A) Always convex (bowl-shaped)
- (B) Always concave
- (C) Non-convex with multiple local minima
- (D) Undefined whenever $m > 2$

Answer: (A)

Solution: J is a sum of squares of an affine function of (w_0, w_1) -- a positive semi-definite quadratic form. Hence it is always convex, guaranteeing that gradient descent converges to the global minimum.

[GATE-style]

Q3. [NAT · 2 marks]

Given data points (1,2), (2,3), (3,5), (4,4), fit $y = w_0 + w_1 x$ by least squares. Find the slope w_1 (correct to 2 decimals).

Answer: 0.80

Solution: $\bar{x}=2.5$, $\bar{y}=3.5$. Deviations $(x-\bar{x}, y-\bar{y})$: $(-1.5, -1.5), (-0.5, -0.5), (0.5, 1.5), (1.5, 0.5)$.
 $\sum (x-\bar{x})(y-\bar{y}) = 2.25 + 0.25 + 0.75 + 0.75 = 4$.
 $\sum (x-\bar{x})^2 = 2.25 + 0.25 + 0.25 + 2.25 = 5$.
 $w_1 = 4/5 = 0.80$.

[GATE-style]

Q4. [NAT · 1 mark]

Using the same data as the previous question, find the intercept w_0 (correct to 2 decimals).

Answer: 1.50

Solution: $w_0 = \bar{y} - w_1 \bar{x} = 3.5 - 0.8 \times 2.5 = 3.5 - 2.0 = 1.50$.

[GATE-style]

Q5. [MCQ · 1 mark]

In gradient descent for linear regression, the update rule is $w := w - \alpha(dJ/dw)$, with $\alpha > 0$. If $dJ/dw > 0$ at the current w , the next update will:

- (A) Increase w
- (B) Decrease w
- (C) Leave w unchanged
- (D) Depends on the sign of α

Answer: (B)

Solution: $w_{\text{new}} = w - \alpha(\text{positive quantity})$, and since $\alpha > 0$, $w_{\text{new}} < w$. So w decreases, moving toward the minimum.

[GATE-style]

Q6. [MSQ · 2 marks]

Which of the following statements about the Mean Squared Error (MSE) cost function used in simple linear regression is/are TRUE?

- (A) It is differentiable everywhere
- (B) It penalizes large errors more heavily than Mean Absolute Error does
- (C) It always has a unique global minimum regardless of the data
- (D) Minimizing it is equivalent to Maximum Likelihood Estimation under a Gaussian noise assumption

Answer: (A), (B), (D)

Solution: (A) MSE is a smooth quadratic -- differentiable everywhere. (B) squaring amplifies large residuals more than $|\text{error}|$ does. (D) is the classical MLE-MSE equivalence for i.i.d. Gaussian noise. (C) is false: e.g. if all x_i are identical, the slope is not uniquely determined.

[GATE-style]

Q7. [NAT · 2 marks]

For the data $(1,2),(2,3),(3,5),(4,4)$, compute the cost $J(w_0, w_1) = (1/2m)\sum(w_0 + w_1x_i - y_i)^2$ at the fitted values $w_0=1.5$, $w_1=0.8$ (correct to 3 decimals).

Answer: 0.225

Solution: Predictions: $x=1 \rightarrow 2.3$ (resid 0.3); $x=2 \rightarrow 3.1$ (resid 0.1); $x=3 \rightarrow 3.9$ (resid -1.1); $x=4 \rightarrow 4.7$ (resid 0.7).
Sum of squares = $0.09 + 0.01 + 1.21 + 0.49 = 1.80$. $J = 1.80/(2 \times 4) = 0.225$.

[GATE-style]

Q8. [MCQ · 1 mark]

If every y_i in a dataset is scaled by a constant c ($y_i' = c \cdot y_i$) while x_i are unchanged, how does the least-squares slope w_1 change?

- (A) $w_1' = w_1$ (unchanged)
- (B) $w_1' = c \cdot w_1$
- (C) $w_1' = w_1/c$
- (D) $w_1' = c^2 w_1$

Answer: (B)

Solution: $w_1 = \sum(x - \bar{x})(y - \bar{y}) / \sum(x - \bar{x})^2$. Scaling y by c scales only the numerator by c , so $w_1' = c \cdot w_1$.

[GATE-style]

Q9. [MCQ · 2 marks]

Consider fitting $y = w_1 x$ (no intercept) to the points (1,1),(2,2),(3,3) by least squares. The minimum value of $J(w_1) = \sum (y_i - w_1 x_i)^2$ is:

- (A) 0
- (B) 1
- (C) 3
- (D) 14

Answer: (A)

Solution: The data exactly satisfies $y=x$, so $w_1=1$ fits every point perfectly, giving zero residuals and $J=0$.

[GATE-style]

Q10. [MCQ · 1 mark]

In simple linear regression, if every data point is duplicated once (so the dataset size doubles but each (x,y) pair now appears twice), the least-squares estimates of w_0 and w_1 are:

- (A) Unchanged
- (B) Doubled
- (C) Halved
- (D) Unpredictable without more information

Answer: (A)

Solution: Duplicating every point scales both the numerator and denominator of the slope/intercept formulas by the same factor (2), and the means $\bar{x}, \bar{y}$ are unaffected, so the estimates stay identical.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 2 marks]

For data points (0,1), (1,3), (2,4), find the least-squares slope w_1 (correct to 2 decimals).

Answer: 1.50

Solution: $\bar{x}=1$, $\bar{y}=8/3 \approx 2.667$. Deviations x: -1,0,1; deviations y: -1.667,0.333,1.333.
Numerator = $1.667+0+1.333 = 3$. Denominator = $1+0+1 = 2$. $w_1 = 3/2 = 1.50$.

[New Practice]

Q12. [NAT · 1 mark]

Using the data of the previous question, find the intercept w_0 (correct to 2 decimals).

Answer: 1.17

Solution: $w_0 = \bar{y} - w_1 \bar{x} = 8/3 - 1.5 \times 1 = 2.667 - 1.5 = 1.17$ (accept 1.16-1.18).

[New Practice]

Q13. [MCQ · 1 mark]

Which quantity is minimized by ordinary least squares in simple linear regression?

- (A) Sum of absolute residuals
- (B) Sum of squared residuals
- (C) Maximum residual
- (D) Sum of residuals

Answer: (B)

Solution: OLS is defined as the estimator minimizing the sum of squared residuals (SSE).

[New Practice]

Q14. [MCQ · 1 mark]

For the cost function $J(w) = w^2 - 4w + 7$, the minimum value and the minimizing w are:

- (A) min = 3 at $w = 2$
- (B) min = 4 at $w = 2$
- (C) min = 3 at $w = 4$
- (D) min = 7 at $w = 0$

Answer: (A)

Solution: $dJ/dw = 2w - 4 = 0 \Rightarrow w = 2$. $J(2) = 4 - 8 + 7 = 3$.

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following statements about the closed-form (normal-equation-style) solution for a **simple** (single-feature) linear regression are TRUE?

- (A) It always requires inverting a large matrix, just as in multi-feature regression
- (B) It exists and is unique whenever the x_i are not all identical
- (C) It can be derived by setting the partial derivatives of the cost function to zero
- (D) It requires iterative parameter updates like gradient descent

Answer: (B), (C)

Solution: For one feature, w_0, w_1 have simple direct formulas (no matrix inversion needed) -- (A) is false. (D) describes gradient descent, not the closed form -- false. (B) and (C) correctly describe the normal-equation approach.

[New Practice]

Q16. [NAT · 1 mark]

For data $y = [2, 4, 6]$ (constant predictor $h(x)=c$), find the value of c that minimizes $\sum(y_i - c)^2$.

Answer: 4

Solution: Minimizing $\sum(y_i - c)^2$ over c gives $c = \text{mean}(y) = (2+4+6)/3 = 4$.

[New Practice]

Q17. [NAT · 1 mark]

Using $c = 4$ from the previous question, find the minimum value of $\sum(y_i - c)^2$.

Answer: 8

Solution: $(2-4)^2 + (4-4)^2 + (6-4)^2 = 4 + 0 + 4 = 8$.

[New Practice]

Q18. [MCQ · 1 mark]

If x and y have a Pearson correlation coefficient of exactly -1 , the least-squares slope w_1 is:

- (A) Positive
- (B) Negative
- (C) Zero
- (D) Undefined

Answer: (B)

Solution: Slope and correlation coefficient always have the same sign; correlation = -1 means a perfect negative slope.

[New Practice]

Q19. [MCQ · 1 mark]

Choosing a learning rate α that is too large in gradient descent for linear regression typically causes:

- (A) Faster guaranteed convergence
- (B) The cost function to oscillate or diverge
- (C) No effect on convergence speed
- (D) w to stay exactly constant

Answer: (B)

Solution: Overly large steps can overshoot the minimum repeatedly, causing $J(w)$ to oscillate or blow up instead of decreasing.

[New Practice]

Q20. [NAT · 2 marks]

Given data (1,4), (2,5), (3,7), (4,10), find the least-squares slope w_1 (correct to 2 decimals).

Answer: 2.00

Solution: $\bar{x}=2.5$, $\bar{y}=6.5$. Deviations x : $-1.5, -0.5, 0.5, 1.5$; deviations y : $-2.5, -1.5, 0.5, 3.5$.

Numerator = $3.75 + 0.75 + 0.25 + 5.25 = 10$. Denominator = $2.25 + 0.25 + 0.25 + 2.25 = 5$. $w_1 = 10/5 = 2.00$.

Topic 2: Ratio and Proportion

Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE 2024, General Aptitude section (common to all GATE papers, including DA), 1 mark]

Q1. [MCQ · 1 mark]

If two distinct non-zero real variables x and y are such that $(x+y)$ is proportional to $(x-y)$, then the value of x/y :

- (A) Depends on both x and y
- (B) Depends only on x
- (C) Depends only on y
- (D) Is a constant, independent of x and y

Answer: (D)

Solution: $(x+y) = k(x-y)$ for some fixed constant $k \Rightarrow x(1-k) = -y(1+k) \Rightarrow x/y = (1+k)/(k-1)$, a fixed number once k is fixed -- it does not depend on the particular x, y chosen.

[VERIFIED PYQ — GATE 2024, General Aptitude section, 1 mark]

Q2. [MCQ · 1 mark]

If $x:y = (a+1):(a-1)$ for real $a > 1$, then $(x^2-y^2) : (x^2+y^2)$ equals:

- (A) $2a : (a^2+1)$
- (B) $(a^2+1) : 2a$
- (C) $2a : (a^2-1)$
- (D) $(a^2-1) : 2a$

Answer: (A)

Solution: Let $x=(a+1)k$, $y=(a-1)k$. $x^2-y^2 = k^2[(a+1)^2-(a-1)^2] = 4ak^2$. $x^2+y^2 = k^2[2a^2+2] = 2k^2(a^2+1)$.
Ratio = $4a : 2(a^2+1) = 2a : (a^2+1)$.

[VERIFIED PYQ — GATE 2024, General Aptitude section, 2 marks]

Q3. [MCQ · 2 marks]

Real variables x, y, z and real constants p, q, r (all denominators non-zero) satisfy $x/(pq-r^2) = y/(qr-p^2) = z/(rp-q^2)$. Which of the following is always TRUE?

- (A) $x+y+z = 0$
- (B) $px+qy+rz = 0$
- (C) $x = y = z$
- (D) $xyz = 0$

Answer: (B)

Solution: Let the common ratio be k , so $x=k(pq-r^2)$, $y=k(qr-p^2)$, $z=k(rp-q^2)$.

$px = k(p^2q-pr^2)$, $qy = k(q^2r-p^2q)$, $rz = k(pr^2-q^2r)$.

Adding: $px+qy+rz = k[(p^2q-pr^2)+(q^2r-p^2q)+(pr^2-q^2r)] = k \cdot 0 = 0$ (every term cancels telescopically).

[GATE-style]

Q4. [NAT · 1 mark]

In a class, the ratio of girls to boys is $3:5$. If there are 24 girls, how many boys are there?

Answer: 40

Solution: girls/boys = $3/5 \Rightarrow 24/\text{boys} = 3/5 \Rightarrow \text{boys} = 24 \times 5/3 = 40$.

[GATE-style]

Q5. [MCQ · 1 mark]

If $a:b = 2:3$ and $b:c = 4:5$, then $a:b:c$ is:

- (A) 8:12:15
- (B) 2:3:5
- (C) 4:6:5
- (D) 8:15:12

Answer: (A)

Solution: Scale $a:b=2:3$ to $8:12$ ($\times 4$); scale $b:c=4:5$ to $12:15$ ($\times 3$) so both share $b=12$. Combine: $a:b:c = 8:12:15$.

[GATE-style]

Q6. [MCQ · 2 marks]

$p:q = 1:2$, $q:r = 4:3$, $r:s = 4:5$. If u is 50% more than s , find $p:u$.

- (A) 16:45
- (B) 8:15
- (C) 45:16
- (D) 15:8

Answer: (A)

Solution: Let $s=15 \Rightarrow r = (4/5) \times 15 = 12 \Rightarrow q = (4/3) \times 12 = 16 \Rightarrow p = q/2 = 8$. $u = 1.5 \times 15 = 22.5$.
 $p:u = 8:22.5 = 16:45$.

[GATE-style]

Q7. [MCQ · 2 marks]

Two quantities are in the ratio $5:7$. If each is increased by 10, the new ratio becomes $3:4$. Find the two original quantities.

- (A) 50 and 70
- (B) 25 and 35
- (C) 45 and 63
- (D) 40 and 56

Answer: (A)

Solution: Let the numbers be $5x, 7x$. $(5x+10)/(7x+10) = 3/4 \Rightarrow 20x+40 = 21x+30 \Rightarrow x = 10$. Numbers = 50, 70.

[GATE-style]

Q8. [MSQ · 2 marks]

If $a:b = c:d$ (all positive), which of the following must also be TRUE?

- (A) $(a+b):(a-b) = (c+d):(c-d)$, provided $a \neq b$
- (B) $a:c = b:d$
- (C) $(a+c):(b+d) = a:b$
- (D) $ab = cd$, always

Answer: (A), (B), (C)

Solution: (A) is the componendo-dividendo identity. (B) is alternendo, valid since $a/b=c/d \Rightarrow a/c=b/d$. (C): if $a/b=c/d=k$, then $a+c=k(b+d)$, so $(a+c)/(b+d)=k=a/b$. (D) is false in general (e.g. $a=1, b=2, c=2, d=4$ gives $a:b=c:d$ but $ab=2 \neq 8=cd$).

[GATE-style]

Q9. [NAT · 1 mark]

The sum of two numbers is 108, and they are in the ratio 5:7. Find the larger number.

Answer: 63

Solution: $5x+7x = 108 \Rightarrow x=9 \Rightarrow \text{larger number} = 7 \times 9 = 63$.

[GATE-style]

Q10. [MCQ · 1 mark]

If $x:y = 3:4$, the value of $(5x+3y) : (5x-3y)$ is:

- (A) 9:1
- (B) 1:9
- (C) 3:1
- (D) 5:1

Answer: (A)

Solution: Take $x=3$, $y=4$: $5x+3y = 15+12 = 27$; $5x-3y = 15-12 = 3$. Ratio = $27:3 = 9:1$.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

A sum of Rs. 720 is divided among A, B, C in the ratio 2:3:4. Find C's share.

Answer: 320

Solution: Total parts = $2+3+4 = 9$. C's share = $(4/9) \times 720 = 320$.

[New Practice]

Q12. [MCQ · 1 mark]

If $(a+b):(a-b) = 5:1$, find $a:b$.

- (A) 3:2
- (B) 2:3
- (C) 5:1
- (D) 1:5

Answer: (A)

Solution: $5(a-b) = a+b \Rightarrow 5a-5b = a+b \Rightarrow 4a = 6b \Rightarrow a/b = 3/2$.

[New Practice]

Q13. [NAT · 2 marks]

The ratio of two numbers is 4:5. If 10 is added to each, the ratio becomes 6:7. Find the smaller original number.

Answer: 20

Solution: Let numbers be $4x$, $5x$. $(4x+10)/(5x+10) = 6/7 \Rightarrow 28x+70 = 30x+60 \Rightarrow x=5$. Smaller number = $4 \times 5 = 20$.

[New Practice]

Q14. [MCQ · 1 mark]

Three numbers are in the ratio 2:3:5, and their sum is 100. Find the largest number.

- (A) 30

- (B) 40
- (C) 50
- (D) 60

Answer: (C)

Solution: Total parts = 10, each part = 10. Largest = $5 \times 10 = 50$.

[New Practice]

Q15. [MSQ · 2 marks]

If b is the mean proportional between a and c (i.e. $a:b = b:c$), which of the following are TRUE?

- (A) $b^2 = ac$
- (B) $b = \sqrt{ac}$
- (C) a, b, c form a geometric progression
- (D) $a+c = 2b$, always

Answer: (A), (B), (C)

Solution: $a/b = b/c \Rightarrow b^2 = ac \Rightarrow b = \sqrt{ac}$ (positive root), and a, b, c is by definition a GP. (D) is the condition for an arithmetic (not geometric) mean, so it is not generally true.

[New Practice]

Q16. [NAT · 1 mark]

Find the mean proportional between 4 and 25.

Answer: 10

Solution: Mean proportional = $\sqrt{4 \times 25} = \sqrt{100} = 10$.

[New Practice]

Q17. [MCQ · 1 mark]

If x, y, z are in continued proportion ($x:y = y:z$) with $x=4$ and $z=9$, find y (positive value).

- (A) 5
- (B) 6
- (C) 7
- (D) 36

Answer: (B)

Solution: $y^2 = xz = 36 \Rightarrow y = 6$.

[New Practice]

Q18. [NAT · 1 mark]

An alloy contains copper and zinc in the ratio 7:3. How many kilograms of zinc are in 50 kg of the alloy?

Answer: 15

Solution: Zinc fraction = $3/10$. Zinc = $(3/10) \times 50 = 15$ kg.

[New Practice]

Q19. [MCQ · 1 mark]

If $a:b = 3:4$ and $b:c = 8:9$, find $a:c$.

- (A) 2:3
- (B) 3:2
- (C) 4:9

(D) 6:9

Answer: (A)

Solution: Scale $a:b = 3:4$ to $6:8$; $b:c = 8:9$ already has $b=8$. So $a:b:c = 6:8:9 \Rightarrow a:c = 6:9 = 2:3$.

[New Practice]

Q20. [NAT · 1 mark]

The three angles of a triangle are in the ratio 2:3:4. Find the largest angle, in degrees.

Answer: 80

Solution: Sum of parts = 9, total angle sum = 180° , so each part = 20° . Largest = $4 \times 20 = 80^\circ$.

Topic 3: Functions, Recursion Basics & Scope Rules

Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE CS 2011, 2 marks (identical concept/format tested under GATE DA's Programming & Data Structures syllabus)]

Q1. [MCQ · 2 marks]

Consider the recursive C function:

```
unsigned int foo(unsigned int n, unsigned int r) {  
    if(n>0) return (n%r) + foo(n/r, r);  
    else return 0;  
}
```

What is the return value of `foo(345, 10)`?

- (A) 345
- (B) 12
- (C) 5
- (D) 3

Answer: (B)

Solution: `foo(345,10) = 5 + foo(34,10)`; `foo(34,10) = 4 + foo(3,10)`; `foo(3,10) = 3 + foo(0,10) = 3+0 = 3`.

So `foo(34,10) = 4+3 = 7`, and `foo(345,10) = 5+7 = 12`. (This computes the digit sum of `n` in base `r`.)

[VERIFIED PYQ — GATE CS, classic recursion/scope question]

Q2. [MCQ · 1 mark]

Which ONE of the following statements about recursion and activation records in the C language is TRUE?

- (A) At most one activation record exists between the current activation record and main's activation record
- (B) The number of activation records between the current record and main's depends on the actual function calling sequence
- (C) The visibility of global variables depends on the actual function calling sequence
- (D) Recursion requires the activation record for the recursive function to be saved on a different stack before the recursive call can be made

Answer: (B)

Solution: (A) is false -- nesting can be arbitrarily deep. (B) is true -- depth depends on how deep the recursion/call chain goes. (C) is false -- C uses static (lexical) scoping, so global visibility does not depend on the call sequence. (D) is false -- all activation records, recursive or not, share the same run-time call stack.

[VERIFIED PYQ — GATE CSE 2024, Set 2]

Q3. [NAT · 2 marks]

Consider the C program:

```
int funcp(){ static int x = 1; x++; return x; }  
int main(){ int x, y; x = funcp(); y = funcp() + x; printf("%d\n", (x+y)); return 0;  
}
```

What value is printed?

Answer: 7

Solution: The static variable inside funcp() is initialized only once, and retains its value between calls.
1st call (x = funcp()): static x: 1→2, returns 2. So main's x = 2.
2nd call (inside y = funcp()+x): static x: 2→3, returns 3. y = 3 + 2 = 5.
Printed value = x+y = 2+5 = 7.

[VERIFIED PYQ — GATE CS, classic, 2 marks]

Q4. [MCQ · 2 marks]

Consider the C function:

```
void reverse(void){ int c; if((c=getchar()) != '\n') reverse(); putchar(c); }
```

Calling reverse() with the input "GATE" followed by a newline will print:

- (A) GATE
- (B) GATEG
- (C) An infinite loop
- (D) ETAG

Answer: (D)

Solution: Each recursive call reads one character before doing anything else, so by the time the base case (newline) is hit, all characters are sitting on the call stack in order G,A,T,E. As the recursive calls return (LIFO order), putchar(c) prints them in reverse: E, T, A, G → "ETAG".

[GATE-style]

Q5. [MCQ · 1 mark]

In C, if a function's local variable has the same name as a global variable, then inside that function:

- (A) Both variables are accessible using a scope-resolution operator, as in C++
- (B) The local variable shadows (hides) the global variable
- (C) It causes a compile-time error
- (D) The global variable is automatically renamed

Answer: (B)

Solution: C has no scope-resolution operator (unlike C++); the innermost (local) declaration always shadows the outer one.

[GATE-style]

Q6. [MCQ · 1 mark]

A recursive function that never reaches a valid base case will typically result in:

- (A) A compile-time error
- (B) A run-time stack overflow
- (C) Silently returning 0
- (D) An infinite loop with no memory issue

Answer: (B)

Solution: Each unfinished call pushes a new activation record onto the call stack; without termination the stack keeps growing until available memory is exhausted, causing a stack overflow.

[GATE-style]

Q7. [NAT · 1 mark]

Consider `int fact(int n){ if(n<=1) return 1; return n*fact(n-1); }`. What value does fact(5) return?

Answer: 120

Solution: $\text{fact}(5) = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

[GATE-style]

Q8. [MCQ · 1 mark]

A variable declared as static inside a C function:

- (A) Is re-initialized to its initial value on every call
- (B) Retains its value between successive calls to the function
- (C) Cannot be modified inside the function
- (D) Is automatically visible in every other source file

Answer: (B)

Solution: Static local variables are initialized only once and persist for the lifetime of the program, retaining their value across calls.

[GATE-style]

Q9. [MSQ · 2 marks]

Which of the following are TRUE about scope and lifetime of variables in C?

- (A) A variable declared inside a block has scope limited to that block
- (B) Global variables have a lifetime equal to the entire program's execution
- (C) Local (automatic) variables are, by default, allocated on the heap
- (D) Parameters passed by value create local copies inside the called function

Answer: (A), (B), (D)

Solution: (C) is false -- ordinary automatic local variables live on the run-time stack, not the heap.

[GATE-style]

Q10. [NAT · 2 marks]

Consider `int f(int n){ if(n==0) return 0; return n + f(n-1); }`. What value does `f(6)` return?

Answer: 21

Solution: $f(6) = 6+5+4+3+2+1+0 = 21$.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 2 marks]

Consider `int g(int n){ if(n<=0) return 1; return n*g(n-2); }`. What value does `g(6)` return?

Answer: 48

Solution: $g(6) = 6 \times g(4) = 6 \times 4 \times g(2) = 6 \times 4 \times 2 \times g(0) = 6 \times 4 \times 2 \times 1 = 48$.

[New Practice]

Q12. [MCQ · 1 mark]

In recursion, the purpose of the "base case" is to:

- (A) Increase the recursion depth
- (B) Stop further recursive calls and begin returning values
- (C) Automatically optimize memory usage
- (D) Convert the recursion into a loop

Answer: (B)

Solution: The base case is the terminating condition; without it recursive calls would never stop unwinding.

[New Practice]

Q13. [MCQ · 1 mark]

Which of the following correctly describes tail recursion?

- (A) The recursive call is the last operation performed in the function, with no pending work after it returns
- (B) The recursive call happens before any other statement in the function
- (C) The function calls itself exactly twice
- (D) The function has no return statement

Answer: (A)

Solution: In tail recursion, nothing remains to be done after the recursive call returns, which allows some compilers to optimize it into a loop.

[New Practice]

Q14. [NAT · 2 marks]

Consider `int sod(int n){ if(n==0) return 0; return n%10 + sod(n/10); }`. What value does `sod(4527)` return?

Answer: 18

Solution: $\text{sod}(4527) = 7 + \text{sod}(452) = 7 + 2 + \text{sod}(45) = 7+2+5+\text{sod}(4) = 7+2+5+4+\text{sod}(0) = 7+2+5+4+0 = 18$ (sum of digits of 4527).

[New Practice]

Q15. [MCQ · 1 mark]

A function that calls itself directly is called ____, whereas a function that calls another function which eventually calls the first one back is called ____.

- (A) Direct recursion; indirect (mutual) recursion
- (B) Indirect recursion; direct recursion
- (C) Tail recursion; head recursion
- (D) Linear recursion; tree recursion

Answer: (A)

Solution: Direct recursion: `f` calls `f`. Indirect/mutual recursion: `f` calls `g`, which calls `f`.

[New Practice]

Q16. [MSQ · 2 marks]

Which of the following statements comparing recursion and iteration are generally TRUE?

- (A) Every recursive function can be converted to an equivalent iterative one using an explicit stack
- (B) Recursive solutions always run faster than iterative ones
- (C) Recursion typically incurs function-call overhead (stack frames) that a simple loop does not
- (D) Recursive functions never terminate

Answer: (A), (C)

Solution: (B) and (D) are false in general -- recursion is often slower due to call overhead, and well-designed recursive functions do terminate via their base case.

[New Practice]

Q17. [NAT · 2 marks]

Consider the Fibonacci-style function `int f(int n){ if(n<=1) return n; return f(n-1)+f(n-2); }` with `f(0)=0`, `f(1)=1`. What value does `f(6)` return?

Answer: 8

Solution: `f`: 0,1,1,2,3,5,8 for `n=0..6`. So `f(6) = 8`.

[New Practice]

Q18. [MCQ · 1 mark]

In C, parameters are passed to functions by which mechanism, by default?

- (A) Call by reference
- (B) Call by value
- (C) Call by name
- (D) Call by address, always

Answer: (B)

Solution: C passes arguments by value by default; call-by-reference-like behavior is only simulated by explicitly passing pointers.

[New Practice]

Q19. [MCQ · 1 mark]

Regarding scope in nested blocks (C-like languages), which statement is TRUE?

- (A) An inner block can access variables declared in an enclosing outer block, if not shadowed
- (B) An outer block can access variables declared inside an inner block after that block ends
- (C) Variables declared in an inner block persist after the block exits
- (D) Scope rules do not apply to loop counter variables

Answer: (A)

Solution: Lexical scoping allows inner blocks to see outer-block variables (unless a same-named local variable shadows it); the reverse is not true.

[New Practice]

Q20. [NAT · 1 mark]

Consider `int f(int n){ if(n==0) return 1; return 2*f(n-1); }`. What value does `f(5)` return?

Answer: 32

Solution: `f(0)=1`, `f(1)=2`, `f(2)=4`, `f(3)=8`, `f(4)=16`, `f(5)=32` (computes 2^n).

Topic 4: String Manipulation with Two-Pointer Technique

Special note for this topic: GATE is a non-coding written exam and has never asked candidates to implement a "two-pointer" algorithm from scratch (that is a coding-interview/judge format, not a GATE format). What GATE does test -- and what genuinely appears in GATE CS papers, in the exact style DA also uses -- is tracing a given C code snippet or analyzing the time/space complexity of a described algorithm. Every question below is therefore written in that authentic GATE code-tracing / complexity-analysis style, and is honestly labelled [GATE-style] throughout, since no verified GATE DA PYQ exists on this narrow a sub-topic.

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [MCQ · 1 mark]

A two-pointer algorithm checks whether a string of length n is a palindrome using pointers $i=0$ and $j=n-1$, comparing $s[i]==s[j]$ and moving $i++$, $j--$ until $i \geq j$. What is its time complexity?

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n^2)$

Answer: (C)

Solution: The pointers move toward each other and together traverse the string once, giving $O(n)$.

[GATE-style]

Q2. [NAT · 2 marks]

Consider:

```
int i=0, j=strlen(s)-1, count=0;
while(i<j){ if(s[i]==s[j]) count++; i++; j--; }
```

For $s = \text{"abcba"}$ (length 5), what is the final value of count?

Answer: 2

Solution: $i=0, j=4$: $s[0]='a'==s[4]='a' \Rightarrow \text{count}=1$. $i=1, j=3$: $s[1]='b'==s[3]='b' \Rightarrow \text{count}=2$. $i=2, j=2$: loop condition $i < j$ fails, stop. Final count = 2.

[GATE-style]

Q3. [MCQ · 1 mark]

In the two-pointer technique for reversing a string in place (swap $s[i]$ and $s[j]$, then $i++$, $j--$), the loop terminates when:

- (A) $i == 0$
- (B) $j == 0$
- (C) $i \geq j$
- (D) $i == j$ only, never $i > j$

Answer: (C)

Solution: Pointers moving toward each other meet or cross exactly when $i \geq j$, which is the correct stopping condition.

[GATE-style]

Q4. [MSQ · 2 marks]

Which of the following problems can be solved in $O(n)$ using the two-pointer technique (on a sorted array or a single string)?

- (A) Checking if a string is a palindrome
- (B) Finding a pair of elements in a sorted array that sum to a target value
- (C) Finding the shortest substring containing all characters of another string (sliding window)
- (D) Sorting an unsorted array

Answer: (A), (B), (C)

Solution: (D) is false -- comparison-based sorting requires $\Omega(n \log n)$ in general; it cannot be done by a simple two-pointer scan.

[GATE-style]

Q5. [NAT · 2 marks]

Given the sorted array [1,3,4,6,8,11] and target sum 10, apply the standard two-pointer method (left=0, right=n-1; if sum<target, left++; if sum>target, right--; else stop). How many sum-computation iterations occur before the pair is found?

Answer: 5

Solution: Iter1: (1,11)=12>10→right--. Iter2: (1,8)=9<10→left++. Iter3: (3,8)=11>10→right--. Iter4: (3,6)=9<10→left++. Iter5: (4,6)=10 → found. Total = 5 iterations.

[GATE-style]

Q6. [MCQ · 1 mark]

The two-pointer technique for merging two sorted arrays of sizes m and n into one sorted array has time complexity:

- (A) $O(m+n)$
- (B) $O(mn)$
- (C) $O(m \log n)$
- (D) $O((m+n) \log(m+n))$

Answer: (A)

Solution: Each pointer scans its own array once; total work is proportional to $m+n$.

[GATE-style]

Q7. [MCQ · 2 marks]

A two-pointer scheme removes duplicates from a sorted array in place. For arr = [1,1,2,2,2,3,4,4], what is the length of the array after removing duplicates in place?

- (A) 3
- (B) 4
- (C) 5
- (D) 8

Answer: (B)

Solution: The distinct elements are 1, 2, 3, 4 -- so the resulting length is 4.

[GATE-style]

Q8. [NAT · 2 marks]

The standard in-place duplicate-removal scheme uses a slow pointer i (initially 0) and a fast pointer j (from 1 to n-1): whenever $\text{arr}[j] \neq \text{arr}[i]$, i is incremented and $\text{arr}[i]$ is set to $\text{arr}[j]$. For arr = [1,1,2,2,2,3,4,4] (n=8, 0-indexed), what is the final (0-indexed) value of i?

Answer: 3

Solution: There are 4 distinct values, occupying indices 0..3 after compaction, so the final i (0-indexed, pointing to the last unique element) equals $4-1 = 3$.

[GATE-style]

Q9. [MCQ · 1 mark]

What auxiliary space complexity does the standard in-place two-pointer approach for palindrome-checking or duplicate-removal typically require (excluding the input itself)?

- (A) $O(1)$
- (B) $O(n)$
- (C) $O(\log n)$
- (D) $O(n^2)$

Answer: (A)

Solution: Only a fixed number of index variables are used, regardless of input size -- constant extra space.

[GATE-style]

Q10. [MCQ · 2 marks]

A sliding-window (two-pointer) check for whether some window of string s (length n) is an anagram of string t (length k), using a fixed-size frequency array over a constant alphabet, has overall time complexity:

- (A) $O(n \times k)$
- (B) $O(n+k)$
- (C) $O(n^2)$
- (D) $O(k^2)$

Answer: (B)

Solution: Each character enters and leaves the sliding window at most once ($O(1)$ amortized work per step), giving $O(n)$, plus $O(k)$ to build the initial target frequency table $\Rightarrow O(n+k)$ overall.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

For $s = \text{"racecar"}$ (length 7), the two-pointer palindrome check starts at $i=0$, $j=6$ and stops when $i \geq j$. How many index pairs (i,j) are actually compared?

Answer: 3

Solution: Pairs compared: $(0,6)$, $(1,5)$, $(2,4)$. Next would be $i=3, j=3$ where $i < j$ fails, so the loop stops without comparing that pair. Total compared = 3.

[New Practice]

Q12. [MCQ · 1 mark]

A two-pointer method to find the longest substring without repeating characters typically works by:

- (A) Expanding a right pointer to grow the window, and moving a left pointer forward to shrink it whenever a repeated character is found (sliding window)
- (B) Moving two pointers toward each other from both ends of the string
- (C) Using binary search combined with recursion
- (D) Sorting the characters of the string first

Answer: (A)

Solution: This is the classic sliding-window pattern -- both pointers only move rightward, never backward.

[New Practice]

Q13. [NAT · 2 marks]

For $s = \text{"abcabcbb"}$, using the sliding-window two-pointer approach, what is the length of the longest substring without repeating characters?

Answer: 3

Solution: The longest such substring is "abc" (length 3); every longer window contains a repeated character.

[New Practice]

Q14. [MCQ · 1 mark]

The "container with most water" style two-pointer algorithm (pointers start at both ends of an array of n heights; the pointer at the shorter height is moved inward at each step) has time complexity:

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(n^2)$
- (D) $O(2^n)$

Answer: (A)

Solution: Each step moves one of the two pointers strictly inward, so there are at most $n-1$ steps total -- $O(n)$.

[New Practice]

Q15. [MSQ · 2 marks]

Which precondition(s) are required for the standard converging two-pointer method to correctly find a pair summing to a target value?

- (A) The array must be sorted
- (B) The array must contain only positive integers
- (C) The array must be non-empty
- (D) All elements must be unique

Answer: (A), (C)

Solution: Sortedness is essential for the left/right pointer movement logic to be correct; a non-empty array is needed for the pointers to be initialized meaningfully. Negative numbers and duplicates are handled fine.

[New Practice]

Q16. [NAT · 2 marks]

Given the sorted array $[2, 5, 7, 9, 13]$ and target 16, apply the two-pointer method ($\text{left}=0, \text{right}=4$). At which iteration number (1-indexed) is the pair found?

Answer: 4

Solution: Iter1: $(2, 13) = 15 < 16 \rightarrow \text{left}++$. Iter2: $(5, 13) = 18 > 16 \rightarrow \text{right}--$. Iter3: $(5, 9) = 14 < 16 \rightarrow \text{left}++$. Iter4: $(7, 9) = 16 \rightarrow \text{found}$. Answer: iteration 4.

[New Practice]

Q17. [MCQ · 1 mark]

The two-pointer technique for "pair with given sum" in a sorted array is an $O(n)$ optimization over which brute-force approach?

- (A) An $O(n)$ single pass
- (B) An $O(n^2)$ check of all pairs
- (C) An $O(\log n)$ binary search only
- (D) An $O(1)$ hashing approach

Answer: (B)

Solution: The naive approach checks every pair of indices (i, j) , costing $O(n^2)$; the two-pointer method reduces this to $O(n)$ using sortedness.

[New Practice]

Q18. [MCQ · 1 mark]

Reversing the order of words in a sentence in place (e.g. "the sky is blue" → "blue is sky the") using two-pointer string reversal is typically done by:

- (A) Reversing the entire string first, then reversing each individual word in place
- (B) Using recursion only, with no pointers involved
- (C) Sorting the words alphabetically
- (D) Scanning with a single pointer moving left to right only

Answer: (A)

Solution: This two-step reversal trick (reverse whole string, then reverse each word) achieves the word-order reversal in $O(n)$ time and $O(1)$ extra space.

[New Practice]

Q19. [NAT · 1 mark]

An in-place two-pointer swap-based reversal is applied to the 8-character string "GATE2026". How many swap operations are performed in total?

Answer: 4

Solution: For a string of length n , the two-pointer reversal performs $n/2$ swaps: $8/2 = 4$.

[New Practice]

Q20. [MCQ · 2 marks]

A sliding-window (two-pointer) approach finds the minimum-length subarray with sum $\geq$ a target S , in an array of n positive integers. Its overall time complexity is:

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(n^2)$
- (D) $O(2^n)$

Answer: (A)

Solution: Each of the two pointers (window start and end) advances forward at most n times in total across the whole run (amortized analysis), giving overall linear time.

Topic 5: Problem Formulation & State-Space Representation in AI

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style, format inspired by the GATE DA 2024 Official Sample Question Paper's A* search item -- reconstructed here as a fully self-contained example since the original figure could not be reliably reproduced]

Q1. [MCQ · 2 marks]

A search graph has states S, A, B, G, where G is the only goal. Edges and costs: $S \rightarrow A$ (2), $S \rightarrow B$ (4), $A \rightarrow G$ (5), $B \rightarrow G$ (1). Heuristic estimates: $h(S)=6$, $h(A)=4$, $h(B)=2$, $h(G)=0$. Using A* **graph** search ($f(n)=g(n)+h(n)$, expand lowest f , break ties alphabetically), via which path is G reached, and with what total cost?

- (A) $S \rightarrow A \rightarrow G$, cost 7
- (B) $S \rightarrow B \rightarrow G$, cost 5
- (C) $S \rightarrow A \rightarrow G$, cost 5
- (D) $S \rightarrow B \rightarrow G$, cost 7

Answer: (B)

Solution: Expand S: generate A ($g=2, h=4, f=6$) and B ($g=4, h=2, f=6$) -- tie broken alphabetically, so A is expanded next, but B remains in the frontier.

Expand A ($f=6$): generate G via A ($g=7, h=0, f=7$).

Frontier now: B ($f=6$), G-via-A ($f=7$). Expand B (lower f): generate G via B ($g=5, h=0, f=5$).

Now the lowest- f node is G-via-B ($f=5$) < G-via-A ($f=7$); expanding it returns the goal via path $S \rightarrow B \rightarrow G$ with total cost 5.

[GATE-style]

Q2. [MCQ · 1 mark]

In the standard formulation of a search problem in AI, which of the following is **NOT** a required component?

- (A) Initial state
- (B) Successor function (actions / transition model)
- (C) Goal test
- (D) A pre-computed optimal solution path

Answer: (D)

Solution: The solution path is what search is meant to find -- it is the output, not a required input component of the problem formulation.

[GATE-style]

Q3. [MCQ · 1 mark]

For the 8-puzzle problem, a "state" in the state space is best represented as:

- (A) The sequence of moves made so far
- (B) The current arrangement (configuration) of the 8 tiles and the blank on the 3×3 grid
- (C) The heuristic value of the current configuration
- (D) The goal configuration only

Answer: (B)

Solution: A state must capture everything needed to determine future moves -- here, the full tile arrangement.

[GATE-style]

Q4. [NAT · 2 marks]

In the 8-puzzle, ignoring reachability restrictions, how many distinct states exist in the raw state space (all permutations of 9 positions: 8 numbered tiles + 1 blank)?

Answer: 362880

Solution: Total states = $9! = 362880$ (only half of these, 181440, are actually reachable from a given start, but the question asks for the raw permutation count).

[GATE-style]

Q5. [MCQ · 1 mark]

The branching factor of a search tree is defined as:

- (A) The total number of nodes in the tree
- (B) The average number of successors generated per state
- (C) The depth of the shallowest goal node
- (D) The number of goal states

Answer: (B)

Solution: Branching factor b is, by definition, the average (or maximum) number of children each node has.

[GATE-style]

Q6. [MSQ · 2 marks]

Which of the following are essential elements of "problem formulation" for a search-based AI agent?

- (A) Initial state
- (B) Actions available in each state
- (C) Transition model (result of applying an action)
- (D) The specific search algorithm (e.g. BFS vs A*) to use

Answer: (A), (B), (C)

Solution: (D) is a separate, later decision about how to search the already-formulated problem -- it is not part of the formulation itself (which also needs a goal test and path cost function, not listed here).

[GATE-style]

Q7. [MCQ · 2 marks]

If a state can be reached via multiple different paths from the initial state, the state space is best represented as a:

- (A) Tree only, never a graph
- (B) Graph (allowing multiple paths/cycles to a state), not strictly a tree
- (C) Linked list
- (D) Stack

Answer: (B)

Solution: Whenever more than one path can reach the same state, the underlying structure is a graph; treating it as a tree would duplicate states.

[GATE-style]

Q8. [NAT · 2 marks]

In the Missionaries and Cannibals problem (3 missionaries, 3 cannibals, boat carries 1 or 2 people), a state is represented as (m, c, b) : missionaries on the left bank, cannibals on the left bank, and boat

position (0 or 1). If m and c each range over $\{0,1,2,3\}$ and b over $\{0,1\}$, what is the maximum number of raw (m,c,b) combinations, before removing invalid states?

Answer: 32

Solution: 4 (values of m) $\times$ 4 (values of c) $\times$ 2 (values of b) = 32 .

[GATE-style]

Q9. [MCQ · 1 mark]

Which of the following best describes "path cost" in a search problem?

- (A) The heuristic estimate from a state to the goal
- (B) A numeric cost assigned to a path, typically the sum of the costs of the individual actions along it
- (C) The number of nodes expanded during search
- (D) The depth of the goal node in the search tree

Answer: (B)

Solution: Path cost $g(n)$ is the additive cost of actions taken to reach a state, distinct from the heuristic $h(n)$.

[GATE-style]

Q10. [MCQ · 1 mark]

A search problem is said to have a well-defined state-space representation if it specifies:

- (A) Initial state, actions, transition model, goal test, and path cost function
- (B) Only the goal state
- (C) Only the heuristic function
- (D) Only the branching factor

Answer: (A)

Solution: These five components together fully and unambiguously define a classical search problem.

Section B — New Practice Questions (10)

[New Practice]

Q11. [MCQ · 1 mark]

For the Water Jug problem (a 4-litre jug and a 3-litre jug, goal: measure exactly 2 litres), a state is naturally represented as:

- (A) A single number (total water)
- (B) An ordered pair (amount in the 4L jug, amount in the 3L jug)
- (C) The sequence of pour operations performed so far
- (D) A Boolean flag indicating success

Answer: (B)

Solution: The state must capture the current water level in each jug, since that fully determines what actions are available next.

[New Practice]

Q12. [NAT · 1 mark]

For the Water Jug problem above (capacities 4 and 3), how many raw states (x,y) are possible with $0 \leq x \leq 4$ and $0 \leq y \leq 3$, ignoring reachability?

Answer: 20

Solution: Number of choices for $x = 5$ (0 to 4), for $y = 4$ (0 to 3). Total = $5 \times 4 = 20$.

[New Practice]

Q13. [MCQ · 1 mark]

In state-space search, "state-space explosion" refers to:

- (A) A bug that causes incorrect goal tests
- (B) The number of reachable states growing extremely large (often exponentially) as the problem size increases
- (C) The heuristic function returning negative values
- (D) Memory being freed too early during search

Answer: (B)

Solution: It is the well-known phenomenon where, e.g., adding one more tile or one more disc multiplies the number of possible states.

[New Practice]

Q14. [MSQ · 2 marks]

Which of the following heuristic properties are relevant to guaranteeing that A* finds an optimal solution?

- (A) Admissibility -- the heuristic never overestimates the true remaining cost (needed for tree search)
- (B) Consistency (monotonicity) -- needed for optimality guarantees in graph search
- (C) The heuristic must always equal the true remaining cost exactly
- (D) The heuristic must be non-negative

Answer: (A), (B), (D)

Solution: (C) is false and unnecessary -- if h always equalled the true cost, the problem would already be solved; admissibility ($h \leq \text{true cost}$) is enough for tree-search optimality, and consistency is the stronger condition needed for graph-search optimality.

[New Practice]

Q15. [MCQ · 1 mark]

In the N-Queens problem, formulated as incremental search (placing one queen per column), a state at depth k of the search tree represents:

- (A) A complete, randomly filled board
- (B) A partial placement of queens in the first k columns, with no two queens attacking each other
- (C) Only the final solution
- (D) The total count of solutions found so far

Answer: (B)

Solution: Incremental formulations build up a partial, consistent configuration one column (or row) at a time.

[New Practice]

Q16. [NAT · 1 mark]

In the standard 8-puzzle formulation, where a move slides a tile into the adjacent blank, what is the maximum branching factor (maximum number of possible actions from a single state)?

Answer: 4

Solution: The blank has 2 neighbours in a corner, 3 on an edge, and 4 in the centre -- so the maximum possible number of actions is 4.

[New Practice]

Q17. [MCQ · 1 mark]

In the classic road-map path-finding problem (e.g. driving from city Arad to city Bucharest), the "goal test" is typically:

- (A) Checking if the current city has the minimum straight-line distance to Bucharest
- (B) Checking if the current city equals "Bucharest"
- (C) Checking if all cities have been visited
- (D) Checking if the path length exceeds some threshold

Answer: (B)

Solution: The goal test simply checks whether the current state matches the specified goal state.

[New Practice]

Q18. [MCQ · 1 mark]

A production system used to implement a search-based AI agent typically consists of:

- (A) A set of rules, a working memory representing the current state, and a control strategy
- (B) Only a goal state
- (C) Only a heuristic function
- (D) A compiler and a linker

Answer: (A)

Solution: Rules (production rules), working memory (state), and a control strategy (which rule to fire/apply next) are the three classical components of a production system.

[New Practice]

Q19. [NAT · 2 marks]

For an uninformed search tree with a uniform branching factor $b=3$, what is the maximum number of nodes generated at depth $d=4$ alone (i.e. b^d)?

Answer: 81

Solution: $3^4 = 81$.

[New Practice]

Q20. [MCQ · 1 mark]

"State space" in AI search refers to:

- (A) The set of all possible configurations (states) reachable from the initial state via valid actions, together with the transitions between them
- (B) Only the initial and goal states
- (C) The memory consumed by the search algorithm's implementation
- (D) The set of all possible heuristic values

Answer: (A)

Solution: This is the standard definition: states + the transition/action structure connecting them.
